Klausur Analysis 2 05.02.07

Nr. 1

$$\vec{B} = \vec{I} \times \frac{\vec{r}}{r^3} \qquad \vec{r} = (x, y, 0)$$

$$r = (x^2 + y^2)^{\frac{1}{2}}$$

$$\vec{I} = (0, 0, I)$$

$$\frac{1}{r^3} \cdot \vec{r}$$

$$\frac{\vec{r}}{r^3} = \left((x^2+y^2)^{-\frac{3}{2}} x \,,\, (x^2+y^2)^{-\frac{3}{2}} y \,,\, 0 \right)$$

$$\vec{B} = \vec{I} \times \frac{\vec{r}}{r^3} = \left(0 - (x^2+y^2)^{-\frac{3}{2}} y I,\, (x^2+y^2)^{-\frac{3}{2}} x I - 0,\, 0 - 0 \right)$$

$$\Rightarrow \underline{\underline{\vec{B} = \left((x^2+y^2)^{-\frac{3}{2}} y I,\, (x^2+y^2)^{-\frac{3}{2}} x I,\, 0 \right)}}$$

a)

$$\text{rot } \vec{B}(x, y, z) = \vec{\nabla} \times \vec{B} \qquad \vec{\nabla} = (\partial_x, \partial_y, \partial_z)$$

$$= \Big(\partial_y \cdot 0 - \underbrace{\partial_z (x^2+y^2)^{-\frac{3}{2}} x I}_{0},\, \underbrace{\partial_z (x^2+y^2)^{-\frac{3}{2}} y I}_{0} - 0,\, \partial_x (x^2+y^2)^{-\frac{3}{2}} x I - \partial_y (x^2+y^2)^{-\frac{3}{2}} y I \Big)$$

$$= \Big(0,\, 0,\, -\frac{3}{\cancel{2}} (x^2+y^2)^{-\frac{5}{2}} \cancel{2} x^2 I + (x^2+y^2)^{-\frac{3}{2}} I - \left(-\frac{3}{\cancel{2}} (x^2+y^2)^{-\frac{5}{2}} \cancel{2} y^2 I + (x^2+y^2)^{-\frac{3}{2}} I \right) \Big)$$

$$= \Big(0,\, 0,\, -3 \frac{1}{r^5} \cdot x^2 I \cancel{+ \frac{1}{r^3} I} + 3 \frac{1}{r^5} \cdot y^2 I \cancel{- \frac{1}{r^3} I} \Big)$$

$$\underline{\underline{\text{rot } \vec{B} = \left(0,\, 0,\, 3 \frac{1}{r^5} I (-x^2 + y^2) \right)}}$$

7) b)

$$\vec{B} = \left((x^2+y^2)^{-\frac{3}{2}} y I, \; (x^2+y^2)^{-\frac{3}{2}} x I, \; 0 \right)$$

$$\operatorname{div} \vec{B} = \partial_x B_x + \partial_y B_y + 0$$

$$\partial_x B_x = -\frac{3}{\cancel{2}} (x^2+y^2)^{-\frac{5}{2}} \cancel{2} x y I = -3 \frac{1}{r^5} x y I$$

$$\partial_y B_y = -\frac{3}{\cancel{2}} (x^2+y^2)^{-\frac{5}{2}} \cancel{2} y x I = -3 \frac{1}{r^5} x y I$$

$$\underline{\underline{\operatorname{div} \vec{B} = -6 I \frac{xy}{r^5}}}$$

c) Nein! Da $\vec{B}$ ein Vektor ist, ist ein Gradient <u>nicht</u> bildbar.

Gradienten lassen sich nur auf Skalarfunktionen anwenden!

Nr. 2

a)

$\text{rot}\ \vec{F} = ?$ $\qquad \vec{F} = (x \sin z,\ z \ln y,\ y)$

$$\text{rot}\ \vec{F} = \vec{\nabla} \times \vec{F}$$

$$= (\partial_x, \partial_y, \partial_z) \times (x \sin z,\ z \ln y,\ y)$$

$$= (\partial_y y - \partial_z z \ln y,\ \partial_z x \sin z - \underbrace{\partial_x y}_{=0},\ \underbrace{\partial_x z \ln y}_{=0} - \underbrace{\partial_y x \sin z}_{=0})$$

$$\underline{\underline{\text{rot}\ \vec{F} = (1 - \ln y,\ x \cos z,\ 0)}}$$

b)

$\text{grad}\ \underbrace{\vec{A} \cdot \vec{r}} = ?$ $\qquad \vec{A} = (a, b, c)$ $\qquad \vec{r} = (x, y, z)$

Skalarprodukt bilden:

$$\vec{A} \cdot \vec{r} = ax + by + cz = F$$

$$\text{grad}\ \vec{A} \cdot \vec{r} = (\partial_x F,\ \partial_y F,\ \partial_z F)$$

$$\underline{\underline{\text{grad}\ \vec{A} \cdot \vec{r} = (a,\ b,\ c)}}$$

2.) c)

div grad U $\qquad U = \frac{Q}{r} \qquad r = \left(x^2+y^2+z^2\right)^{\frac{1}{2}}$

$$U = Q \cdot \left(x^2+y^2+z^2\right)^{-\frac{1}{2}}$$

$$\text{grad } U = Q\left(-\cancel{\tfrac{1}{2}}\left(x^2+y^2+z^2\right)^{-\frac{3}{2}} \cdot \cancel{2}x,\ -\cancel{\tfrac{1}{2}}\left(x^2+y^2+z^2\right)^{-\frac{3}{2}} \cancel{2}y,\ -\cancel{\tfrac{1}{2}}\left(x^2+y^2+z^2\right)^{-\frac{3}{2}} \cdot \cancel{2}z\right)$$

$$\text{grad } U = -Q\left(x^2+y^2+z^2\right)^{-\frac{3}{2}} \cdot (x, y, z)$$

$$\underline{\text{grad } U = -\frac{Q}{r^3}(x, y, z)}$$

$$\text{div grad } U = \left(\partial_x \text{grad } U_x + \partial_y \text{grad } U_y + \partial_z \text{grad } U_z\right)$$

$$\partial_x \text{grad } U_x = -Q\left(-\cancel{\tfrac{3}{2}}\left(x^2+y^2+z^2\right)^{-\frac{5}{2}} \cdot \cancel{2}x^2 + \left(x^2+y^2+z^2\right)^{-\frac{3}{2}} \cdot 1\right)$$

$$\partial_x \text{grad } U_x = -Q\left(-3\ \frac{1}{r^5}\ x^2 + \frac{1}{r^3}\right)$$

$$\partial_y \text{grad } U_y = -Q\left(-3\ \frac{1}{r^5}\ y^2 + \frac{1}{r^3}\right)$$

$$\partial_z \text{grad } U_z = -Q\left(-3\ \frac{1}{r^5}\ z^2 + \frac{1}{r^3}\right)$$

$$\text{div grad } U = -Q\left(-3\ \frac{1}{r^5}\underbrace{\left(x^2+y^2+z^2\right)}_{r^2} + 3\ \frac{1}{r^3}\right)$$

$$= -Q\left(-3\ \frac{1}{r^3} + 3\ \frac{1}{r^3}\right)$$

$$\underline{\underline{\text{div grad } U = 0}}$$

nur 70 Punkte ?

o.o

$y'' + 4y' + 5y = 5x^2 - 32x + 5$ ⇐ zu lösende DGL

Hom.:

$\lambda^2 + 4\lambda + 5 = 0 \qquad \lambda_{1/2} = -2 \pm \sqrt{4-5} \qquad \lambda_{1/2} = -2 \pm j$

$$\underline{y_H = e^{-2x}\left[C_1 \sin(x) + C_2 \cos(x)\right]}$$

Inh.

$\Rightarrow y_{inh.} = a + bx + cx^2 \qquad y'_{inh.} = b + 2cx \qquad y''_{inh} = 2c$

Einsetzen in DGL

$2c + 4b + 8cx + 5a + 5bx + 5cx^2 = 5x^2 - 32x + 5$

$5cx^2 + (8c + 5b)x + (2c + 4b + 5a) = 5x^2 - 32x + 5$

$\Rightarrow 5c = 5 \quad |:5 \qquad \Rightarrow 8 \cdot 1 + 5b = -32 \quad |-8$

$\underline{c = 1} \qquad 5b = -40 \quad |:5$

$\underline{b = -8}$

$\Rightarrow 2 \cdot 1 + 4 \cdot (-8) + 5a = 5 \quad |+30$

$5a = 35 \quad |:5$

$\underline{a = 7}$

$\Rightarrow \underline{y_{inh} = 7 - 8x + x^2}$

$$y = y_H + y_{inh} = \underline{\underline{e^{-2x}\left[C_1 \sin(x) + C_2 \cos(x)\right] + 7 - 8x + x^2}}$$